

# Circuit

MP

Teach You Physics Hand In Hand

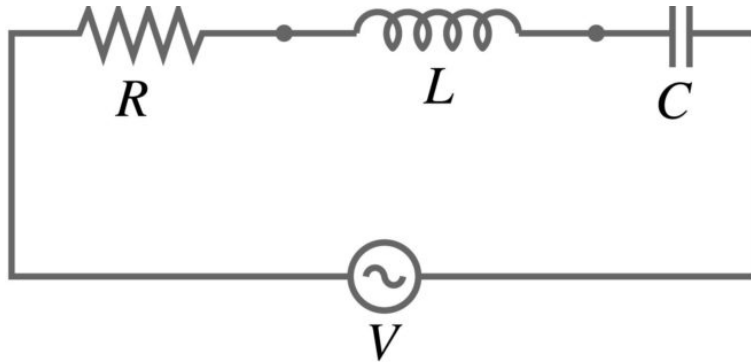
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## 1 SOLUTION to Problem in the Wave post

- Level of difficulty of the problem: **2/10**
- Waves are in phase  $\Rightarrow$  must be sine function without phase difference.
- Opposite propagation direction of  $-\hat{x}$ , so  $kx + wt$ .
- $\pi$  phase shift of the  $\vec{E}$ -field and  $-\hat{x}$  propagation direction, so  $\vec{B}$  along  $-\hat{z}$ .
- $\Rightarrow \boxed{\vec{B}(x, t) = -\hat{z}B_0 \sin(kx + wt)}$  or  $\boxed{\vec{B}(x, t) = -\hat{z}\frac{E_0}{c} \sin(kx + wt)}$

## 2 Circuit Components

- The solenoid inserted into a circuit is called an **inductor**.
- The conducting plates inserted into a circuit are called a **capacitor**.



- where  $L$  is **self inductance** (measures the ability to produce magnetic flux) and  $C$  is **capacitance** (measures the ability to store charges).
- Voltage dissipation in the inductor  $= L \frac{dI}{dt}$ , that in the capacitor  $= \frac{Q}{C}$ .

### 3 Computing the Current in a.c. R-L-C Circuit

Using **Complex Impedance**  $\tilde{Z}$  where  $i = \sqrt{-1}$ :

$$\tilde{Z} = |\tilde{Z}|e^{i \arctan\left(\frac{wL - \frac{1}{wC}}{R}\right)} = e^{i \arctan\left(\frac{wL - \frac{1}{wC}}{R}\right)} \sqrt{R^2 + \left(wL - \frac{1}{wC}\right)^2}$$

Using **Complex Voltage**  $\tilde{\mathcal{E}}(t) = \mathcal{E}_0 e^{iwt}$ :

$$\begin{aligned} \tilde{I}(t) &= \frac{\mathcal{E}_0}{\sqrt{R^2 + \left(wL - \frac{1}{wC}\right)^2}} e^{iwt} e^{-i \arctan\left(\frac{wL - \frac{1}{wC}}{R}\right)} \\ \Rightarrow I(t) &= \frac{\mathcal{E}_0}{\sqrt{R^2 + \left(wL - \frac{1}{wC}\right)^2}} \cos \left[ wt - \arctan \left( \frac{wL - \frac{1}{wC}}{R} \right) \right] \end{aligned}$$

### 4 Computing the Current in d.c. R-L-C Circuit

By **Kirchoff's Rules** (total voltage supplied = total voltage dissipated):

$$\begin{aligned} \frac{d^2}{dt^2} Q(t) + \frac{R}{L} \frac{d}{dt} Q(t) + \frac{1}{LC} Q(t) - \frac{\mathcal{E}(t)}{L} &= 0 \\ \Rightarrow Q(t) &= A e^{-\frac{Rt}{2L}} \cos \left( \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t + \phi \right) + Q(t=0) \\ \Rightarrow I(t) &= \frac{A}{\sqrt{LC}} e^{-\frac{Rt}{2L}} \cos \left[ \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} t + \phi + \pi - \arctan \left( \frac{2L}{R} \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \right) \right] \end{aligned}$$

where  $A$  and  $\phi$  are some constants determined by initial conditions.

### 5 Transformer and Mutual Inductance

**Mutual inductance**  $M$  measures the ability of a primary inductor to induce current in a secondary inductor. Define  $MI_p = \Phi_B$ , where  $\Phi_B$  is magnetic flux through secondary inductor and  $I_p$  is primary current. By Faraday's Law and Lenz's Law, voltage dissipation in the transformer is given by:

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{dMI_{other}}{dt} = -M \frac{d}{dt} I_{other}$$

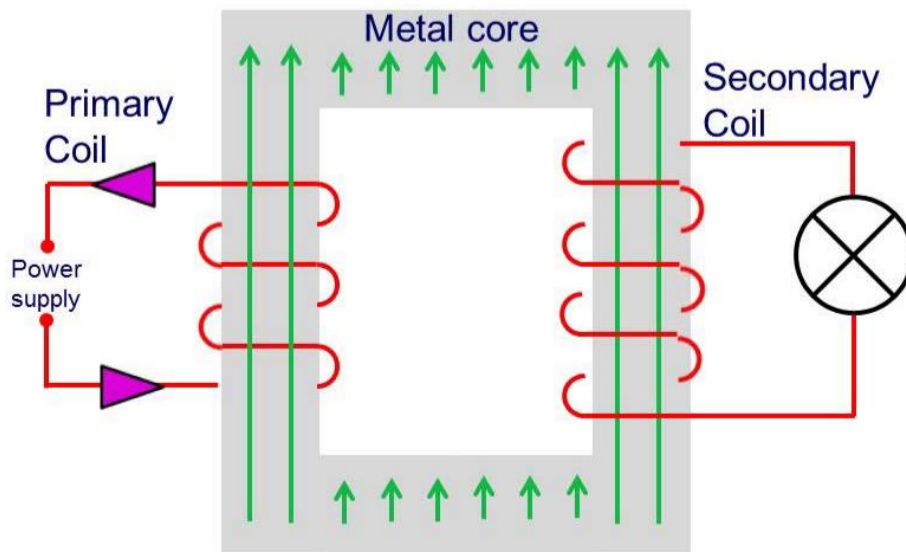
Please make sure that you know this, as it is **frequently** asked in DSE.

## 6 Problem

[20 points] A typical transformer circuit is composed of an a.c. source with sinusoidal voltage  $V(t) = V_0 \cos(\omega t)$ , an ideal transformer, and a light bulb with resistance  $R$ . Inductors have mutual inductance  $M$  and equal self inductance  $L$ .

a) [10 points] What is the sinusoidal **secondary current**  $I_s$ ? (Hint: It should be something like  $I_0 \cos(\omega t)$ , where  $I_0$  is something you should find.)

b) [10 points] What is the **average power**  $\langle P \rangle$  delivered to the light bulb?



Ignore the field lines and the number of coils in the graph. Please note that you CANNOT directly use  $V_p I_p = V_s I_s$  as  $I_p, I_s, V_s$  are unknowns.

The solution will be uploaded in the next post and MP love you all :)

Finally I know how to highlight in latex hahaha